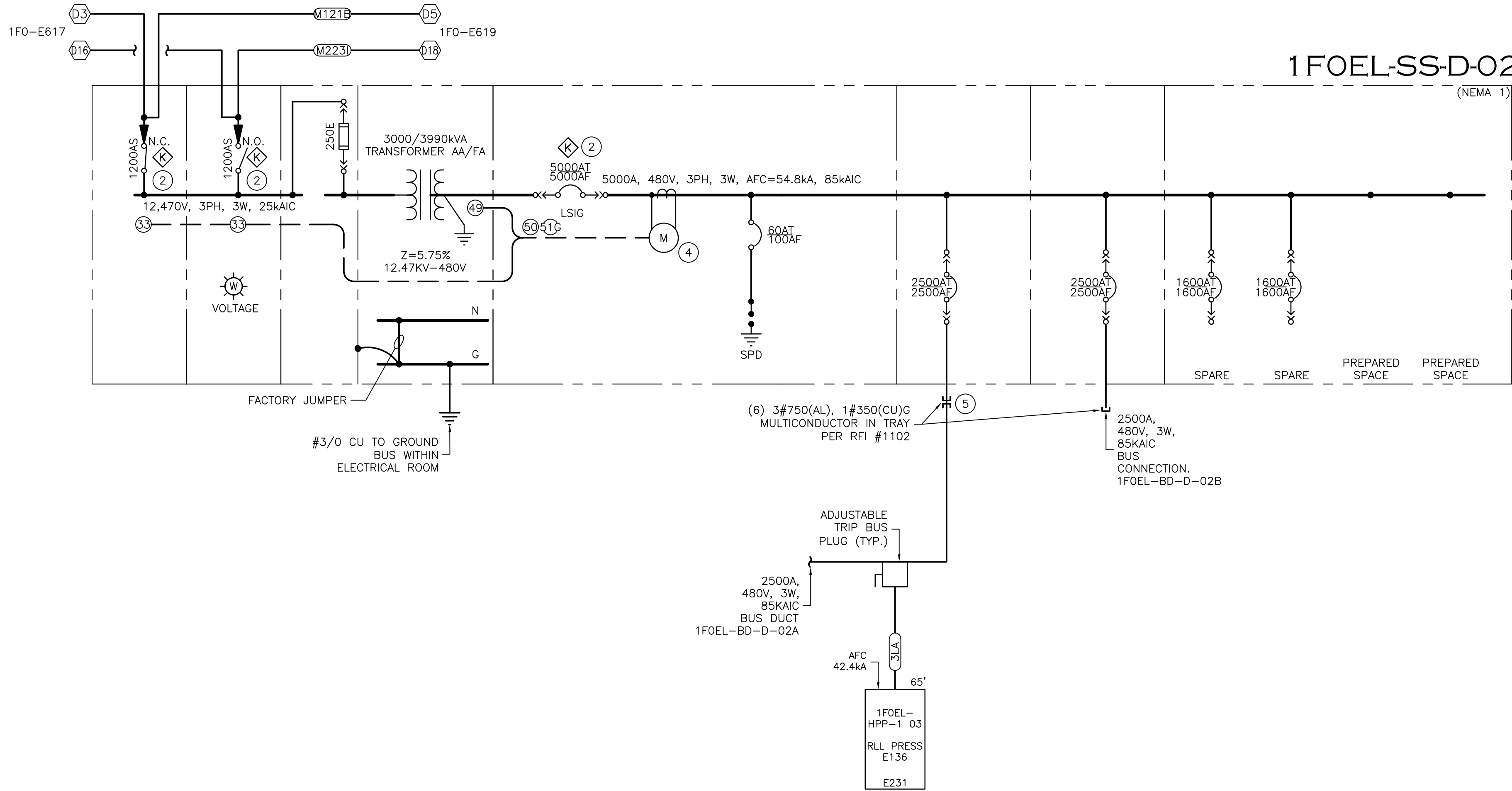


GENERAL NOTES

- A. REFERENCE FEEDER SCHEDULE, E011
- B. CABLE LENGTHS WHEN INDICATED ON ONE-LINE ARE APPROXIMATE AND ARE FOR REFERENCE ONLY FOR CALCULATIONS AND ARE NOT TO BE USED FOR MATERIAL TAKEOFFS.
- C. SECONDARY UNIT SUBSTATIONS: REFER TO SPECIFICATION #26 11 16 12.
- D. UNIT SUBSTATION SCADA COMMUNICATIONS SHALL INCLUDE ALL HARDWARE DEVICES TO COMMUNICATE I/O (VIA MODBUS TCP COMMUNICATIONS) INCLUDING, BUT NOT LIMITED TO, THE FOLLOWING:
PRIMARY SWITCH STATUS (OPEN/CLOSED)
TRANSFORMER WINDING TEMPERATURE
TRANSFORMER FAN RUN STATUS
BREAKER STATUS
POWER METER DATA
- E. ALL SWITCHBOARDS, SWITCHGEAR, AND PANELBOARDS SUPPLIED BY FEEDERS SHALL BE PERMANENTLY MARKED TO INDICATE EACH DEVICE OR EQUIPMENT WHERE THE POWER ORIGINATES. THE LABEL SHALL BE PERMANENTLY AFFIXED, OF SUFFICIENT DURABILITY TO WITHSTAND THE ENVIRONMENT INVOLVED, AND NOT HANDWRITTEN.

KEYED NOTES

1. NOT USED
2. PRIOR TO CLOSING A SWITCH, THE OTHER SWITCH MUST BE OPENED USING KIRK KEY INTERLOCK TO PREVENT CONNECTION OF MULTIPLE SOURCES. REFER TO SEQUENCE OF OPERATIONS BY MANUFACTURER IN EQUIPMENT SUBMITTAL.
3. NOT USED.
4. SQUARE D/ SCHNEIDER ION 9000 WITH RS-485 COMMUNICATIONS.
5. ENGINEERED CABLE BUS ASSEMBLY, CONDUIT, OR CABLE TRAY TO BUS DUCT CABLE TAP BOX.



ONE LINE DIAGRAM - (1FO) ELECTRODE "D" (PARTIAL)
NO SCALE

BUS DUCT NAME										
1FOEL-BD-D-02A										
BUS PLUG	EQUIPMENT	TAG #	RATINGS		LOADING (KVA)				FLA	NOTES
			VOLTS	AMP	AMP	LOAD	DEM	DEM		
				FRAME	TRIP			LOAD		
COLD OIL CONTROLLER PANEL (#2)		2	480	150	40	22	0.4	9	27	
COLD HOT OIL CONTROLLER PANEL		3	480	150	100	76	0.4	31	92	
COLD OIL CONTROL PANEL		4	480	150	70	35	0.4	14	42	
COLD OIL CONTROLLER PANEL (#1)		5	480	150	40	22	0.4	9	27	
IHU PANEL		6	480	150	100	59	0.4	24	71	
COLD OIL CONTROLLER PANEL (#2)		8	480	150	40	22	0.4	9	27	
COLD HOT OIL CONTROLLER PANEL		9	480	150	100	76	0.4	31	92	
COLD OIL CONTROL PANEL		10	480	150	70	35	0.4	14	42	
COLD OIL CONTROLLER PANEL (#1)		11	480	150	40	22	0.4	9	27	
IHU PANEL		12	480	150	100	59	0.4	24	71	
COLD OIL CONTROLLER PANEL (#2)		14	480	150	40	22	0.4	9	27	
COLD HOT OIL CONTROLLER PANEL		15	480	150	100	76	0.4	31	92	
COLD OIL CONTROL PANEL		16	480	150	70	35	0.4	14	42	
COLD OIL CONTROLLER PANEL (#1)		17	480	150	40	22	0.4	9	27	
IHU PANEL		18	480	150	100	59	0.4	24	71	
COLD OIL CONTROLLER PANEL (#2)		20	480	150	40	22	0.4	9	27	FUTURE
COLD HOT OIL CONTROLLER PANEL		21	480	150	100	76	0.4	31	92	
COLD OIL CONTROL PANEL		22	480	150	70	35	0.4	14	42	
COLD OIL CONTROLLER PANEL (#1)		23	480	150	40	22	0.4	9	27	FUTURE
IHU PANEL		24	480	150	100	59	0.4	24	71	
COLD OIL CONTROLLER PANEL (#2)		26	480	150	40	22	0.4	9	27	FUTURE
COLD HOT OIL CONTROLLER PANEL		27	480	150	100	76	0.4	31	92	
COLD OIL CONTROL PANEL		28	480	150	70	35	0.4	14	42	
COLD OIL CONTROLLER PANEL (#1)		29	480	150	40	22	0.4	9	27	FUTURE
IHU PANEL		30	480	150	100	59	0.4	24	71	
COLD OIL CONTROLLER PANEL (#2)		32	480	150	40	22	0.4	9	27	FUTURE
COLD HOT OIL CONTROLLER PANEL		33	480	150	100	76	0.4	31	92	
COLD OIL CONTROL PANEL		34	480	150	70	35	0.4	14	42	
COLD OIL CONTROLLER PANEL (#1)		35	480	150	40	22	0.4	9	27	FUTURE
IHU PANEL		36	480	150	100	59	0.4	24	71	
COLD OIL CONTROLLER PANEL (#2)		38	480	150	40	22	0.4	9	27	FUTURE
COLD HOT OIL CONTROLLER PANEL		39	480	150	100	76	0.4	31	92	
COLD OIL CONTROL PANEL		40	480	150	70	35	0.4	14	42	
COLD OIL CONTROLLER PANEL (#1)		41	480	150	40	22	0.4	9	27	FUTURE
IHU PANEL		42	480	150	100	59	0.4	24	71	
COLD OIL CONTROLLER PANEL (#2)		44	480	150	40	22	0.4	9	27	FUTURE
COLD HOT OIL CONTROLLER PANEL		45	480	150	100	76	0.4	31	92	
COLD OIL CONTROL PANEL		46	480	150	70	35	0.4	14	42	
COLD OIL CONTROLLER PANEL (#1)		47	480	150	40	22	0.4	9	27	FUTURE
IHU PANEL		48	480	150	100	59	0.4	24	71	
1FOEL-HPP-1 03		49	480	400	250	15	1.0	15	18	

NOTES:
*800A AND ABOVE BUS PLUGS REQUIRE MANUFACTURER INSTALLED BUS PLATE. SEE POWER PLAN FOR LOCATION

TOTAL LOAD (KVA) 1737

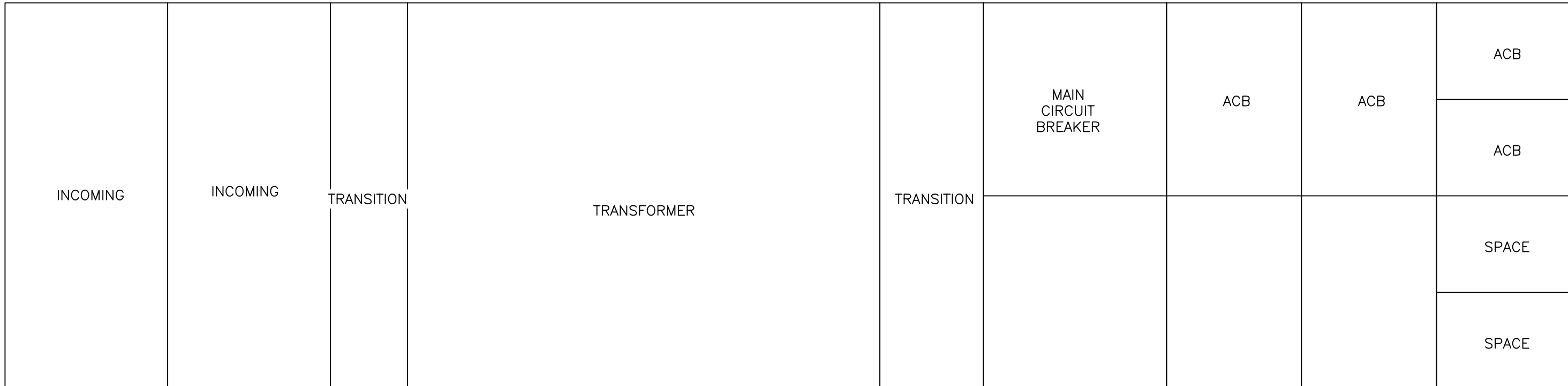
704 TOTAL DEMAND LOAD (KVA)
846.81 DEMAND AMPS
2500 RATING (A)
34 % LOADING

BUS DUCT NAME										
1FOEL-BD-D-02B										
BUS PLUG	EQUIPMENT	TAG #	RATINGS		LOADING (KVA)				FLA	NOTES
			VOLTS	AMP	AMP	LOAD	DEM	DEM		
				FRAME	TRIP			LOAD		
INVERTER PANEL		1	480	600	600	436	0.4	174	524	
COLD HOT OIL CONTROLLER PANEL		2	480	250	100	76	0.4	31	92	
INVERTER PANEL		4	480	600	600	436	0.4	174	524	
COLD HOT OIL CONTROLLER PANEL		5	480	250	100	76	0.4	31	92	
INVERTER PANEL		7	480	600	600	436	0.4	174	524	
COLD HOT OIL CONTROLLER PANEL		8	480	250	100	76	0.4	31	92	
INVERTER PANEL		10	480	600	600	436	0.4	174	524	
COLD HOT OIL CONTROLLER PANEL		11	480	250	100	76	0.4	31	92	
INVERTER PANEL		13	480	600	600	436	0.4	174	524	FUTURE
COLD HOT OIL CONTROLLER PANEL		14	480	250	100	76	0.4	31	92	
INVERTER PANEL		16	480	600	600	436	0.4	174	524	FUTURE
COLD HOT OIL CONTROLLER PANEL		17	480	250	100	76	0.4	31	92	
INVERTER PANEL		19	480	600	600	436	0.4	174	524	FUTURE
COLD HOT OIL CONTROLLER PANEL		20	480	250	100	76	0.4	31	92	
INVERTER PANEL		22	480	600	600	436	0.4	174	524	FUTURE
COLD HOT OIL CONTROLLER PANEL		23	480	250	100	76	0.4	31	92	

NOTES:
*800A AND ABOVE BUS PLUGS REQUIRE MANUFACTURER INSTALLED BUS PLATE. SEE POWER PLAN FOR LOCATION

TOTAL LOAD (KVA) 4097

1639 TOTAL DEMAND LOAD (KVA)
1971.2 DEMAND AMPS
2500 RATING (A)
79 % LOADING



EQUIPMENT ELEVATION
NO SCALE

BUS DUCT LOAD SCHEDULE
NO SCALE

WARE MALCOMB

ARCHITECTURE CIVIL ENGINEERING
PLANNING BRANDING
INTERIORS BUILDING MEASUREMENT

2777 E Camelback Rd. Suite #205
Phoenix, AZ 85016
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10/23/25

30536
WILLIAM J. BETHURUM, V.
SEAL EXPIRES: 09/30/26

VOLTA

1730 E. Northern Avenue, Ste. 100, Phoenix, AZ 85020
Tel: (480) 686-0111 voltaaz.com

REG 215101 10.23.25

PROJECT ALPHA

1F0 - MAIN MANUFACTURING BUILDING
EAST PECOS ROAD
QUEEN CREEK, ARIZONA 85140

ELECTRICAL ONE LINE DIAGRAM

DATE	REMARKS
10/10/2024	DC-074 1F0 CD PROGRESS SET (MEP ONLY)
11/01/2024	DC-074 1F0 FULL CD PACKAGE
12/17/2024	1F0 DC-074 REVISIONS
03/17/2025	1F0 CD REVISIONS
07/24/2025	1F0 CD PERMIT RESUBMISSION
10/23/2025	1F0 BULLETIN #1

PA/PM: WB

DRAWN BY: ETS/AOI/AUG

JOB NO.: PHX21-5038-00

SHEET

1F0-E616

PLOT DATE: 10/23/2025